

MASTER'S THESIS DEFENSE

Enhancing Stability in Direct Training of Spiking Neural Networks

Membrane Potential Initialization & Threshold-Robust Surrogate Gradients

Hyunho Kook (국 현 호)

Advisor: Prof. Eunhyeok Park

Committee: Prof. Jungseul Ok · Prof. Jae-Joon Kim

Why Spiking Neural Networks?

DNNs are powerful but expensive

Training and serving large models routinely consumes megawatt-scale power; even edge inference is hard under tight energy budgets.

SNNs: an event-driven alternative

Information flows as discrete spikes over time. Paired with neuromorphic hardware, computation skips inactive neurons — orders of magnitude lower energy.

But direct training is unstable

Recent direct-training methods narrowed the accuracy gap, yet two persistent instabilities still hold SNNs back.

THIS THESIS

Two instabilities, two principled fixes.

① Temporal Covariate Shift

in the membrane potential — invisible to existing remedies.

② Gradient pathology

when the firing threshold becomes a learnable parameter — never explicitly characterized before.

Spiking Neurons in 60 Seconds

Three discrete-time updates, plus one trick to get gradients through them

Leaky Integrate-and-Fire neuron

Integrate

$$M[t] = (1 - 1/\tau) \cdot U[t-1] + (1/\tau) \cdot I_{in}[t]$$

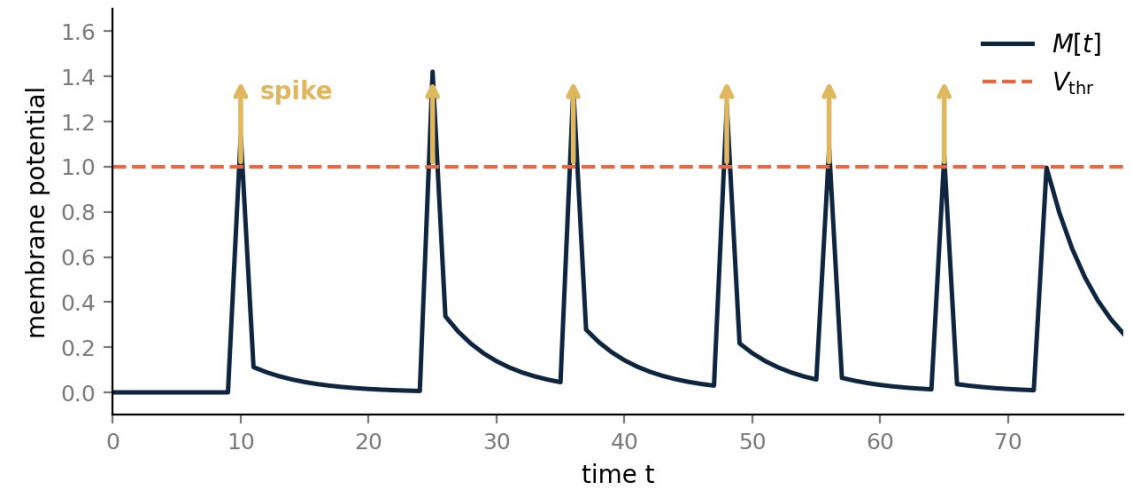
Fire

$$S[t] = H(M[t] - V_{thr}) \in \{0, 1\}$$

Reset

$$U[t] = M[t] - V_{thr} \cdot S[t] \quad (\text{soft reset})$$

Two learnable parameters: τ (leakage) · V_{thr} (threshold).



$M[t]$ integrates inputs, fires at V_{thr} , resets — repeat.

Heaviside H is not differentiable

Surrogate gradient: backward pass uses smooth $f'(x) \approx H'(x)$.

Everything in this thesis turns on $M[t]$ and V_{thr} · TCS at $M[t]$ · pathology at V_{thr}

Surrogate Gradient — Training Through the Spike

Spikes are non-differentiable; smooth surrogates re-open backprop

The problem

Forward: $S = H(M - V_{thr}) \in \{0, 1\}$

Backward: $\partial S / \partial M = \delta(M - V_{thr})$

Heaviside has zero gradient almost everywhere, with a delta spike exactly at threshold — gradient information cannot flow.

The fix: surrogate gradient (SG)

Forward unchanged. In backward, replace δ with smooth $f'(x)$:

$$\partial S / \partial M \approx f'(x), \quad |x| < \gamma/2$$

f' is peak-normalized (max = 1), supported on a window of width γ around 0. Forward stays binary; gradient flows.

Common shapes • same window, different curve

► Rectangular ► Triangular ► Arctan ► Sigmoid

Shape matters less than scale — TrSG works across all four.

γ controls the active window WIDTH; height is set by the chain rule.

Two natural choices for x

$x = M - V_{thr} \rightarrow$ AS-SG (Absolute scale)

$x = M / V_{thr} - 1 \rightarrow$ RS-SG (Relative scale)

Same SG curve, two ways to plug in $V_{thr} \rightarrow$ analyzed in Part II.

Diagnosis ①: TCS Lives in the Membrane Potential

What everyone notices

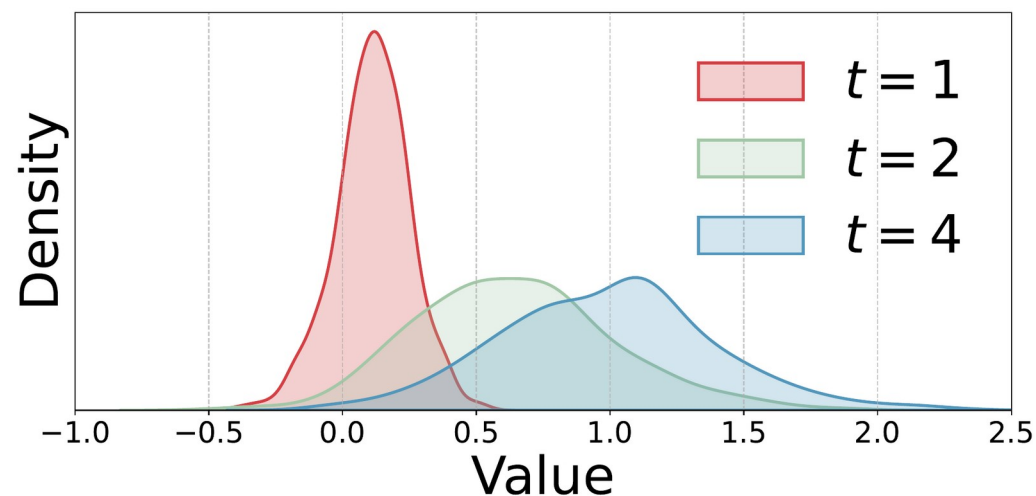
Activation distributions drift across timesteps. Batch-norm statistics estimated at one timestep no longer fit at another — accuracy drops.

What existing fixes do

TEBN, TAB add per-timestep BN parameters.
More compute, more parameters — and they don't reach the cause.

Our claim

TCS originates one level deeper — in $M[t]$ itself.



$M[t]$ under tdbn — distribution clearly drifts as t grows

Diagnosis ②: Trainable V_{thr} Breaks the Gradient

PLIF, LTMD, DIET-SNN all let V_{thr} learn — but the surrogate gradient is scale-sensitive

$V_{thr} \ll 1$ (small threshold)

- ▶ AS-SG floods — fixed window covers almost the entire distribution
- ▶ RS-SG explodes — the $1/V_{thr}$ factor blows the magnitude up
- ▶ Real example: at $V_{thr} = 0.1$, RS-SG diverges to NaN within 100 epochs

$V_{thr} \gg 1$ (large threshold)

- ▶ AS-SG starves — fixed window covers a tiny tail of the distribution
- ▶ RS-SG vanishes — the same $1/V_{thr}$ factor shrinks the magnitude
- ▶ Real example: at $V_{thr} = 2.0$, AS-SG hits 0% active neurons by epoch 100

These are not corner cases.

Trainable V_{thr} in practice drifts well below 1 in early layers and above 2 in deep ones — exactly the regimes where AS-SG and RS-SG both fail. We'll quantify in Part II.

Why Does the Membrane Potential Drift?

Because it's a Markov chain pulled toward a unique distribution

The recurrence is a Markov chain

$U[t+1] = f(U[t], I_{in}[t+1])$ — Markov by construction; i.i.d. inputs $\Rightarrow$ time-homogeneous

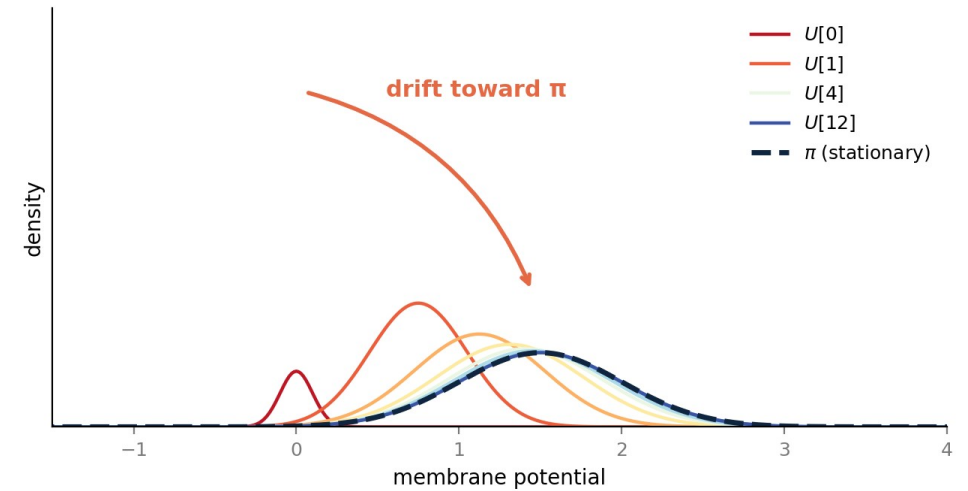
Theorem (Doebelin minorization)

Under (i.i.d. inputs) + (bounded $U[t]$), the chain has a unique stationary π . Moreover,

$$\|P^n(x, \cdot) - \pi(\cdot)\|_{TV} \leq C \cdot (1 - \epsilon)^n$$

Implications

- ▶ Initial state $\neq \pi \Rightarrow$ drift is inevitable
- ▶ Standard $U[0] = 0$ is far from π
- ▶ Don't fix BN — fix the initial state



Distribution of $U[t]$ for $t = 0, 1, 2, 3, 4, 5, 8, 12$

Starting from $U[0]=0$ (sharp red), the distribution exponentially approaches the stationary π (dashed navy).

MP-Init — Initialize at the Stationary Mean

$U[T]$ is the closest sample to $\pi \rightarrow$ use a running mean of $E[U[T]]$

Lemma — best constant is the mean

$$\operatorname{argmin}_c E[(\pi - c)^2] = E[\pi]$$

Practical estimator

$U[T]$ is the closest sample to π . Maintain a per-layer running mean
 $\mu \leftarrow \beta \cdot \mu + (1 - \beta) \cdot \text{mean}(U[T])$ over training, $\beta = 0.9$.
 At every batch, set $U[0] = \mu$.

Algorithm (training-only)

```
init  $\mu_l \leftarrow 0$  for each layer  $l$ 

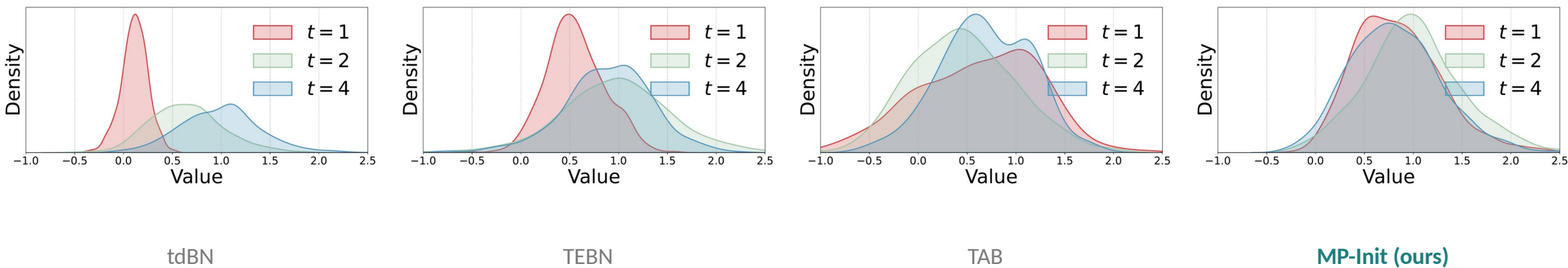
for each batch:
   $U_l[0] \leftarrow \mu_l$ 
  run  $T$  LIF steps as usual
   $m \leftarrow \text{mean}(U_l[T] \text{ of active neurons })$ 
   $\mu_l \leftarrow \beta \cdot \mu_l + (1 - \beta) \cdot m$ 
```

Inference: μ frozen $\rightarrow$ standard LIF dynamics

Zero changes to BN, no per-timestep parameters, full neuromorphic-hardware compatibility.

MP-Init Works — in Both Distribution and Metric

Membrane potential distributions across timesteps (ResNet-19 / CIFAR-100)



Consistent accuracy gain on every baseline

Setting	Baseline	+ MP-Init
CIFAR-100 · T=2 · tdBN	72.35	74.01
CIFAR-100 · T=4 · tdBN	75.55	76.09
CIFAR-100 · T=4 · TEBN	75.96	76.45
CIFAR-100 · T=4 · TAB	76.25	77.24
DVS-CIFAR10 · T=10 · tdBN	76.60	77.37

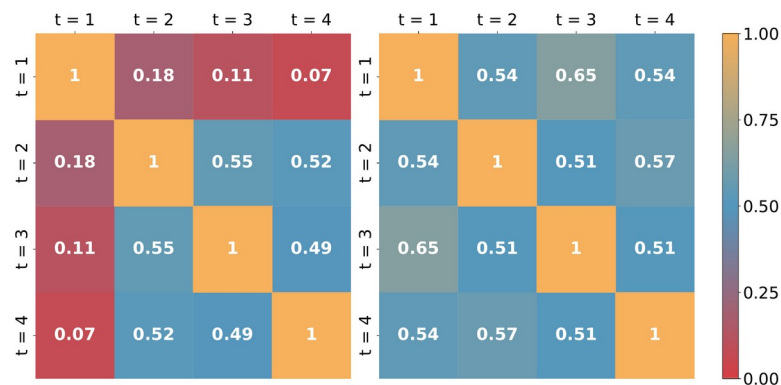
+1.66 %pt at T = 2

Biggest gain in the most latency-critical regime — exactly where SNN energy efficiency matters.

Why MP-Init Works — A Closer Look

Aligning distributions aligns gradients — and improves the firing-rate frontier

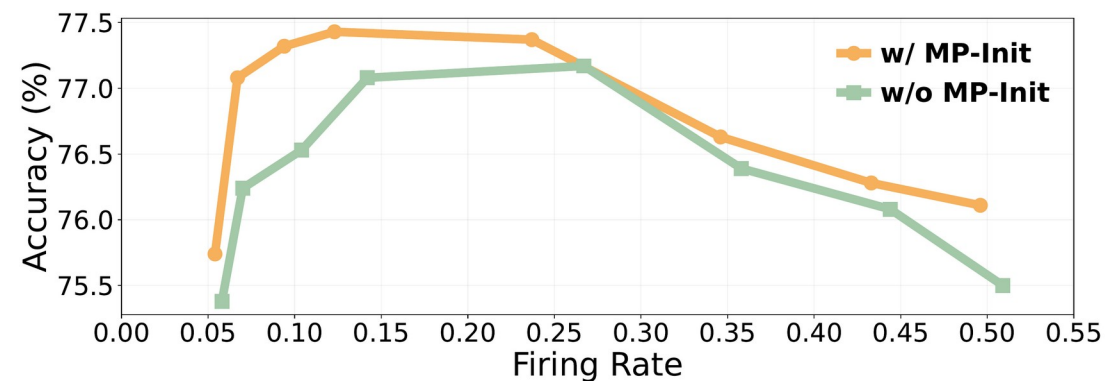
Gradient cosine similarity across timesteps



ResNet-19 • CIFAR-100 • 4 timesteps

- ▶ Without MP-Init: gradients at $t=1$ conflict with later steps (cos 0.07–0.18)
- ▶ With MP-Init: temporal ensemble realigned (cos ≈ 0.5)
- ▶ Stable optimization — not just stable inference

Accuracy vs. firing rate (Pareto frontier)



ResNet-19 • CIFAR-100 with sparsity regularizer

- ▶ MP-Init dominates baseline at every firing rate
- ▶ Even at 6% firing rate, beats all baseline settings
- ▶ Gains come from spiking BETTER, not spiking MORE

Higher accuracy from more stable training — efficiency-preserving improvements, not accuracy-via-spend.

Two Surrogate-Gradient Families

AS-SG and RS-SG — look similar, behave very differently when V_{thr} is trainable

$$\partial S / \partial M \approx f'(x) \cdot \text{rectangular } f' = 1 \text{ on } |x| < \gamma/2 \text{ (peak-normalized)} \cdot \gamma = \text{active window width}$$

AS-SG • Absolute-Scale

$$x = M[t] - V_{thr}$$

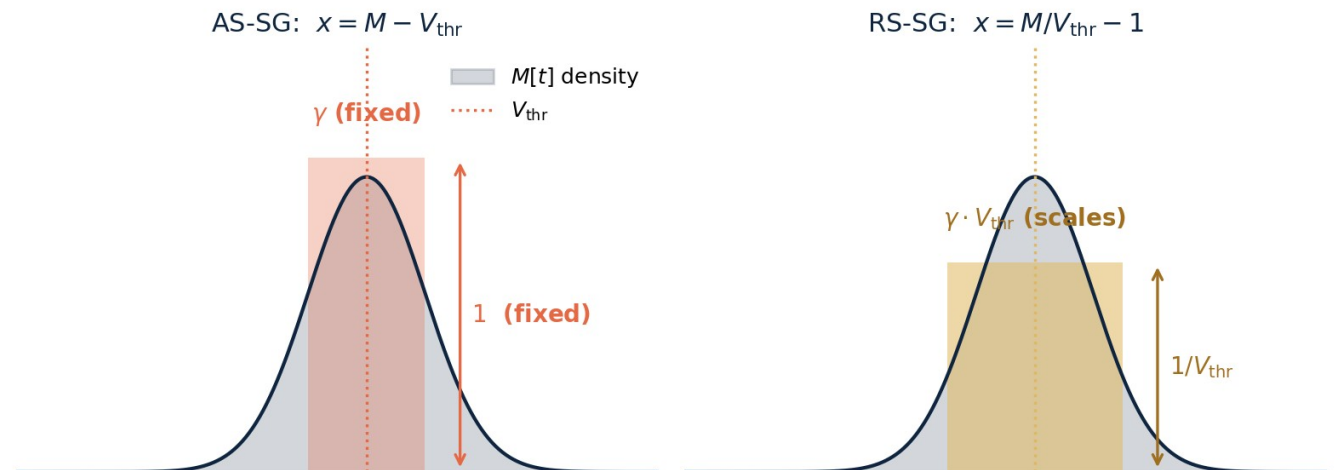
Window of **FIXED** width γ , centered at V_{thr}

RS-SG • Relative-Scale

$$x = M[t] / V_{thr} - 1$$

Window **SCALES** with V_{thr} — but magnitude $/ V_{thr}$

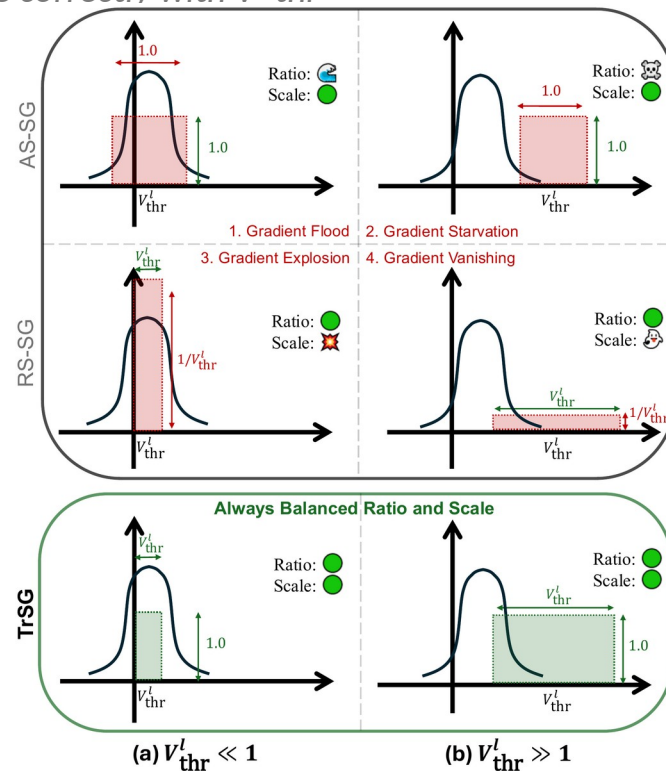
Gradient-window vs membrane distribution



Same when V_{thr} is fixed. Once V_{thr} trains, AS-SG and RS-SG break in opposite ways.

Four Failure Modes — and One Fix

Both window width AND gradient magnitude must scale correctly with V_{thr}



Each colored box = active gradient region. width = window · height = magnitude

✗ Flood

$$AS-SG \cdot V_{thr} \ll 1$$

✗ Starvation

$$AS-SG \cdot V_{thr} \gg 1$$

✗ Explosion

$$RS-SG \cdot V_{thr} \ll 1$$

✗ Vanishing

$$RS-SG \cdot V_{thr} \gg 1$$

✓ Balanced

$$TrSG \cdot \text{all regimes}$$

TrSG — One-Line Fix

Multiply the threshold on the forward path; chain rule cancels the bad factor

The recipe

1 Relative-scale window

$$x = M / V_{\text{thr}} - 1$$

2 Multiply on forward

$$O[t] = V_{\text{thr}} \cdot S[t]$$

3 Chain rule cancels

$$\partial O / \partial M = f'(x)$$

Chain-rule cancellation



Window adapts with V_{thr} · gradient magnitude is threshold-invariant.

Inference Stays Binary

TrSG is a training-only trick — neuromorphic hardware never sees it

Training

Forward: $O[t] = V_{thr} \cdot S[t]$ (real-valued — gradient flows cleanly through V_{thr})

At deployment, absorb V_{thr} into the next layer's weights

$W'_{\{l \rightarrow l+1\}} = V_{thr}^l \cdot W_{\{l \rightarrow l+1\}}$ (one-time rescaling per layer)

Forward at test time: $O[t] = S[t] \in \{0, 1\}$ (standard LIF — back to binary spikes)

Standard LIF operators at inference. Hardware compatibility preserved.

Why TrSG Works — A Stable Gradient

Three stability metrics tell the same story across extreme V_thr

AbsStr

mean |gradient|

low → vanishing, high → exploding

RatioAG

% neurons with active gradient

low → starvation, high → flooding

GradCV

std / mean of |gradient|

high → unstable, hard-to-tune updates

Stress-test metrics at epoch 100 — CIFAR-100, ResNet-19, $\tau=2.0$ frozen

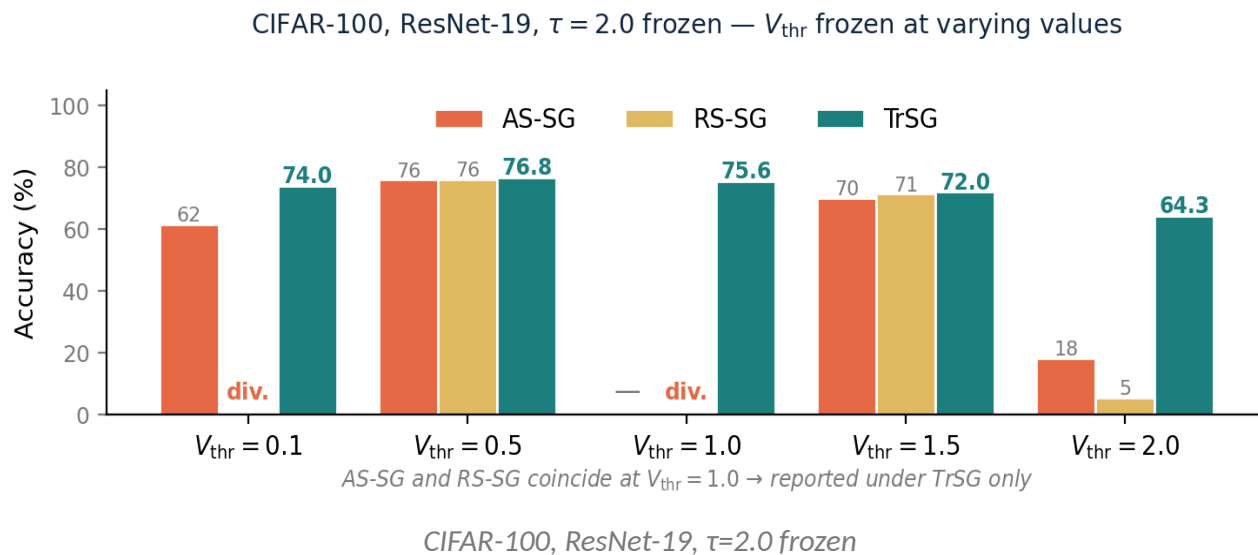
V_thr = 0.1 (small threshold)				V_thr = 2.0 (large threshold)			
SG	AbsStr	RatioAG	GradCV	SG	AbsStr	RatioAG	GradCV
AS-SG	0.0145	90.20%	1.22	AS-SG	0.0000	0.00%	0.000
RS-SG	NaN	0.00%	NaN	RS-SG	0.0021	0.10%	2.25
TrSG	0.0153	21.45%	0.87	TrSG	0.0323	6.90%	0.73

TrSG keeps all three metrics in the healthy range — at every threshold.

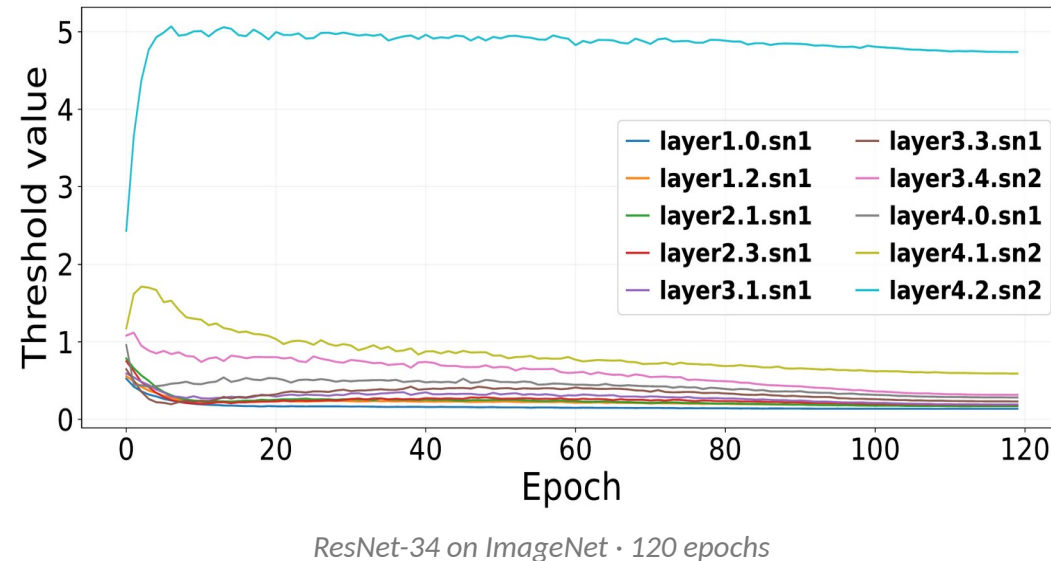
TrSG Robust Across All V_{thr}

Direct accuracy evidence + real trainings actually visit the extreme regimes

Accuracy with V_{thr} frozen at varying values



Trainable V_{thr} trajectories — ImageNet

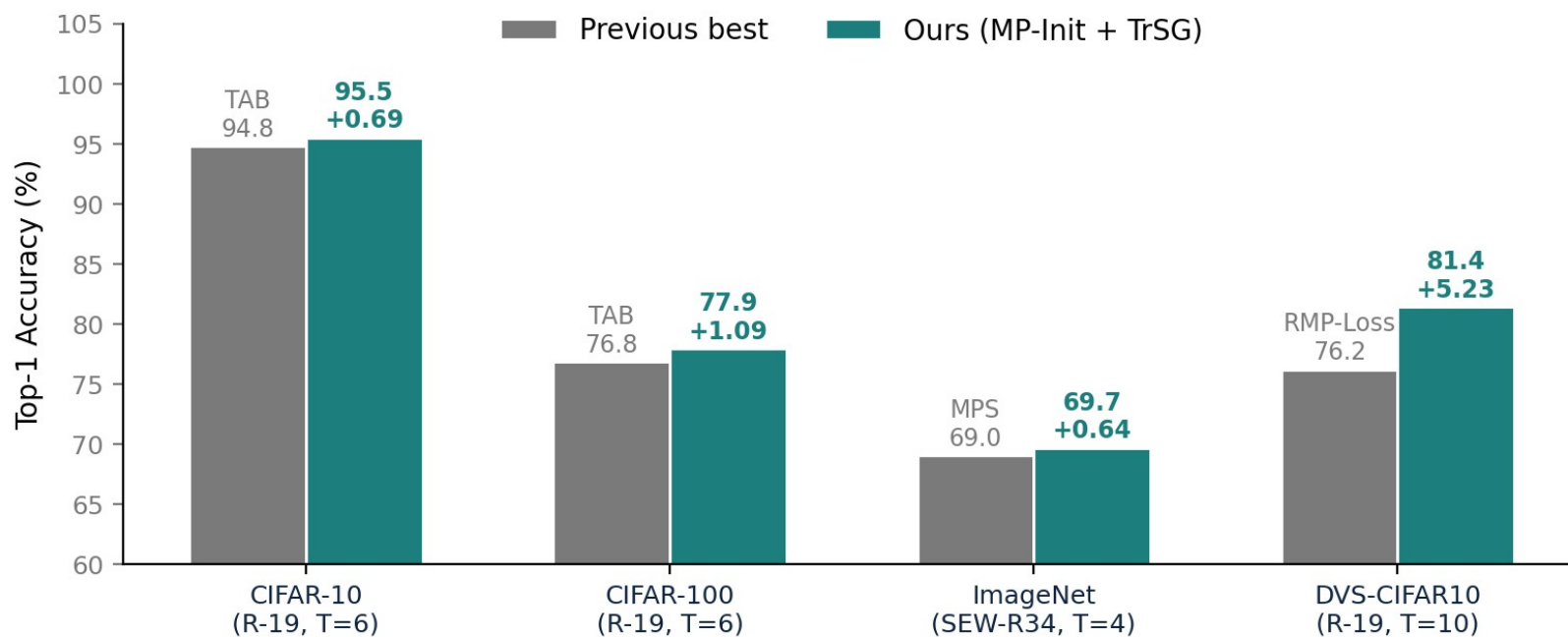


- Frozen V_{thr} accuracy: TrSG wins at every threshold; AS-SG/RS-SG fail outside $V_{thr} \approx 1$
- Real trainings drift $V_{thr} \ll 1$ in early layers and $\gg 1$ in deep layers — exactly the failure regimes

Threshold-robustness is a practical necessity, not a theoretical edge case.

Main Results — SOTA Across the Board

Standard LIF neurons • no AutoAugment / Cutout • three trials



+1.09 %pt

CIFAR-100 • beats TAB

+0.64 %pt

ImageNet • beats MPS

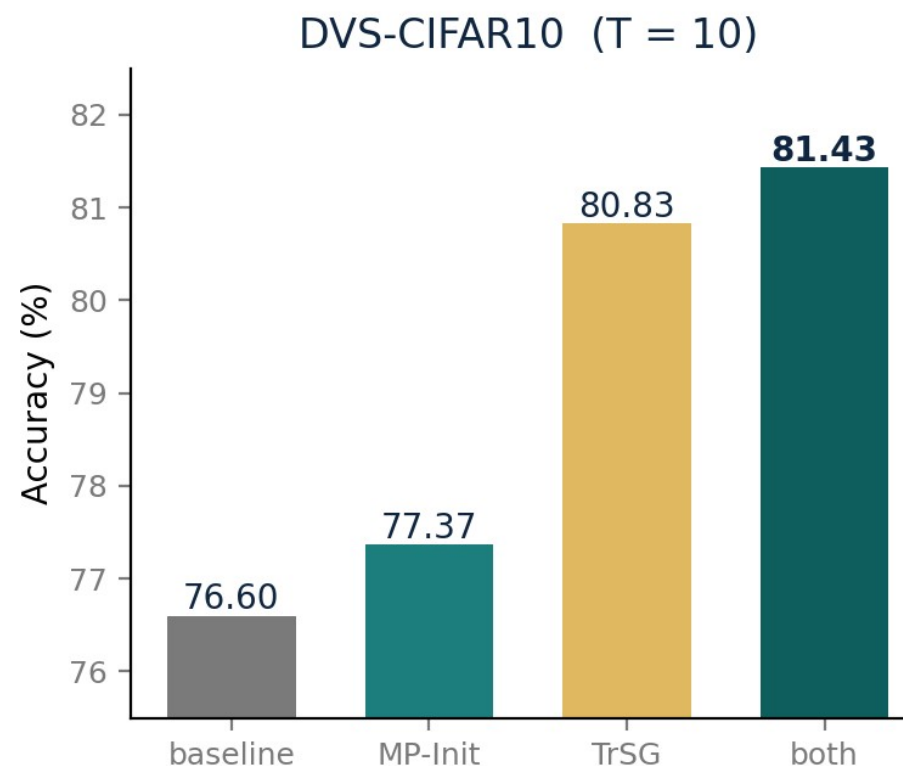
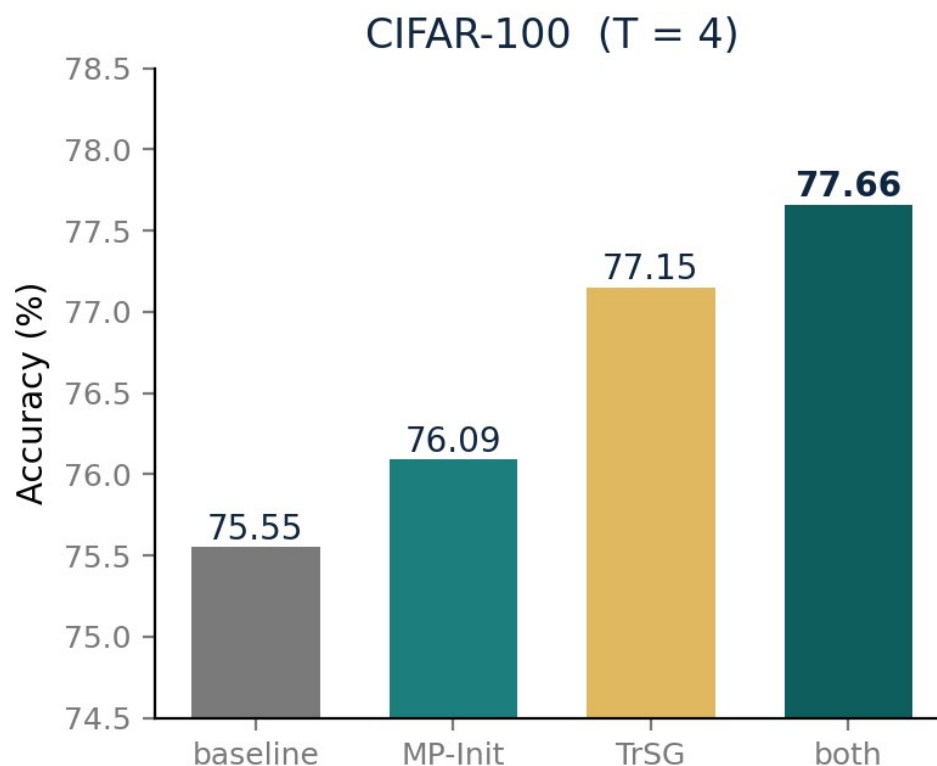
+5.23 %pt

DVS-CIFAR10 • beats RMP-Loss

Biggest gain on the most challenging (event-based) dataset — DVS-CIFAR10 +5.23 %pt

Ablation — Independent and Complementary

Each method helps alone; gains stack additively (largest combined effect on event-based data)



Takeaway

MP-Init alone

+0.5 ~ 0.8

TrSG alone

+1.6 ~ 4.2

Both

+2.1 ~ 4.8

The two fixes are orthogonal — diagnoses aim at different components, not the same problem twice.

Beyond Classification

Transformer SNNs • object detection • energy preserved

Transformer SNN (QKFormer, ImageNet, T=4)

Baseline 78.80 → + ours 79.90 (+1.10 %pt)
AS-SG 77.09 → RS-SG 78.70 → TrSG 79.90
Drop-in SG swap on a state-of-the-art Transformer SNN.

Object detection (COCO-2017, EMS-ResNet10, T=3)

mAP@0.5 0.291 → 0.305 (+0.014)
mAP@0.5:0.95 0.138 → 0.147 (+0.009)
First evidence the principles transfer beyond classification.

Energy under matched configuration (CIFAR-100)

Baseline 1.57 mJ • 75.58 %
+ MP-Init + TrSG 1.67 mJ • 77.68 %
+2.10 %pt at +6 % energy → orders of magnitude better than
spike-rate-multiplying baselines (cf. Pareto frontier on slide 9).

All four surrogate shapes

Rectangular • Triangular • Arctan • Sigmoid
TrSG remains the best across initial V_{thr} , initial τ ,
and surrogate shape — never tied for second.

No architectural changes, no extra losses, no per-timestep parameters — just drop in.

What This Talk Was About

Two long-standing instabilities of direct SNN training, traced and fixed.

Both fixes are minimal, principled, and preserve standard inference.

MP-Init

TCS originates in the membrane potential — fix the initial state.

Per-layer running mean of $U[T]$.

Zero overhead, standard LIF inference.

TrSG

Threshold-induced gradient pathology — fix it with one forward-path multiplication.

Window adapts, magnitude is invariant.

Standard LIF after training.

Validated

SOTA on CIFAR-10/100, ImageNet, DVS-CIFAR10.

Generalizes to Transformer SNNs (QKFormer +1.10) and detection (COCO +0.014 mAP).

All without architectural changes.

CONCLUSION

**Two instabilities, two principled fixes,
one cohesive story.**

LIMITATIONS

- ▶ Per-layer constant approximates π only at the mean — richer parameterizations are open.
- ▶ Hardware with fixed neuron parameters needs quantization-aware adaptation.
- ▶ i.i.d. assumption empirically supported, but tighter bounds welcome.

FUTURE WORK

- ▶ Hardware-aware adaptation: Loihi, Akida, TrueNorth
- ▶ Beyond LIF: adaptive LIF, Izhikevich, foundation-scale SNNs

Thank you. Questions?

BACKUP MATERIAL

Appendix

Q&A reference: hardware, datasets, prior work

- A1** Can SNNs run on real chips?
- A2** Where SNN efficiency comes from
- A3** Energy: SNN chip vs DNN platforms
- A4** What is an event-based dataset?

- A5** From DVS events to SNN input
- A6** How were SNNs trained before?
- A7** Prior work on TCS & threshold learning

Can SNNs Actually Run on Chips?

Production-grade neuromorphic silicon already exists.

Yes — and at scale.

Several research and commercial chips natively run spiking models. Our methods (MP-Init, TrSG) are training-time techniques that produce the same LIF model these chips already accept — so the trained network ports directly without architectural changes.

Key takeaway

Hardware is here. The bottleneck is training quality at low T.

Chip	Year	Process	Capacity	Headline metric
IBM TrueNorth	2014	65 nm	1 M neur. 256 M syn.	70 mW · 26 pJ/event
Intel Loihi	2018	14 nm	130 K neur. 130 M syn.	23.6 pJ / op
Intel Loihi 2	2021	Intel 4	1 M neur. graded spikes	improved · multi-bit
SpiNNaker 2	2024	22 nm	152 K neur. 152 M syn. / chip	ARM-based · digital
Tianjic	2019	28 nm	40 K neur. SNN+ANN hybrid	Tsinghua · hybrid arch.
Speck (SynSense)	2022	65 nm	328 K neur. DVS on-chip	< 10 mW · always-on
BrainScaleS-2	2022	65 nm	512 analog neur.	1000× biological speed

Refs: Merolla 2014 · Davies 2018 · Mayr 2019 · Pei 2019 · SynSense 2022 · BrainScaleS team 2022.

Where SNN Efficiency Comes From

Three structural advantages — none requires a new training scheme.

① Event-driven

No spike → no compute. Inactive neurons skip MAC and memory access entirely.

Typical activity: 5–20 % per timestep

② Multiplication-free

Spike $\in \{0, 1\} \rightarrow$ W-spike degenerates to W or 0. Each synapse becomes an Accumulate (AC), not a MAC.

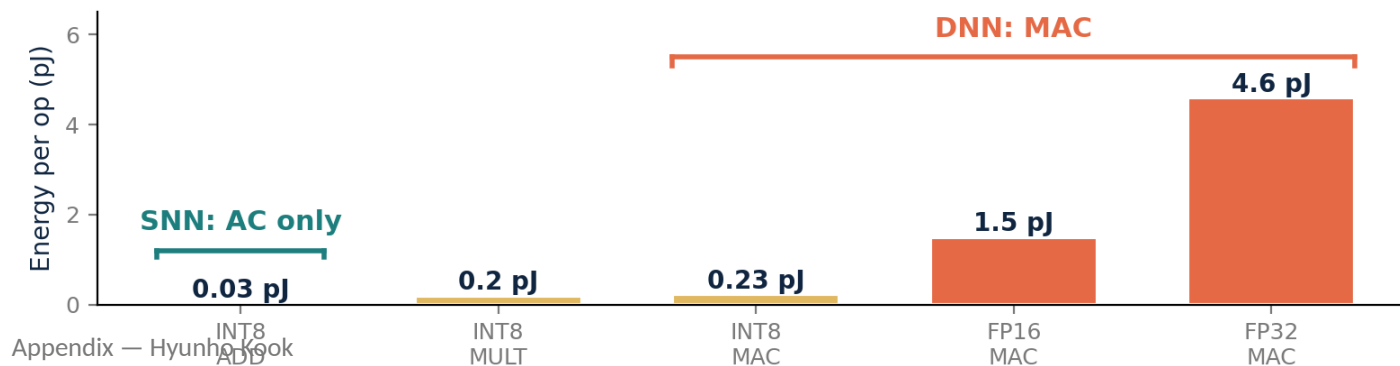
INT8 AC $\approx$ 8× cheaper than INT8 MAC (45 nm)

③ Co-located memory

Neuromorphic cores keep synaptic weights in local SRAM, on-chip mesh routing — no DRAM trip.

DRAM 64-bit $\approx$ 1300 pJ vs SRAM $\approx$ 5 pJ

Per-operation energy at 45nm [Horowitz, ISSCC 2014]



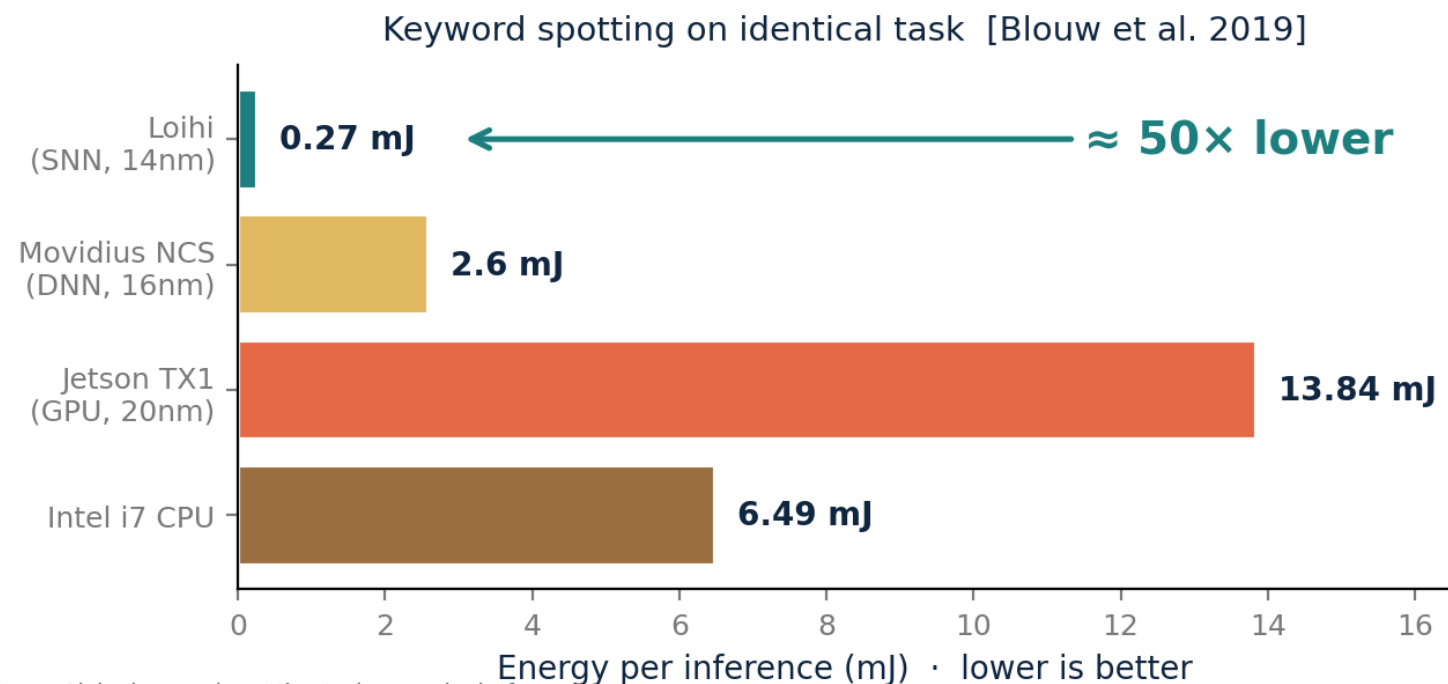
Bottom line

Efficiency is structural — comes from sparsity + binary spikes + co-located memory.

MP-Init / TrSG protect (1) by stabilizing low-T training.

Energy: SNN Chip vs DNN Platforms

Same task, four hardware platforms — what does the gap actually look like?



Same Aloha keyword-spotting task on each platform.

Sources: Blouw et al. 2019 (keyword-spotting benchmark) ·
Davies et al. 2021 (LASSO 48× vs CPU) · Horowitz ISSCC 2014 (per-op energy).

How big is the gap?

Sparse / event tasks

~50–100× vs embedded GPU
(keyword spot, gesture, sparse coding)

Static images (CIFAR / ImageNet)

~2–5× vs GPU at iso-accuracy
(per published SNN/CNN comparisons)

Why so task-dependent?

Energy $\propto T \times \text{spike-rate} \times \text{neurons}$.
Low-T + high-sparsity → big gap.

Why this matters for our work

MP-Init enables $T = 2$ SOTA → directly multiplies energy savings.

What Is an Event-Based Dataset?

Output of asynchronous DVS cameras — sparse, fast, high dynamic range.

DVS: Dynamic Vision Sensor

Each pixel asynchronously emits an event when its log-intensity changes by a threshold.

Output: a stream of (x, y, t, polarity) tuples.

μs

temporal resolution

>120 dB

dynamic range

Sparse

only changes emit

No blur

no exposure window

In this thesis

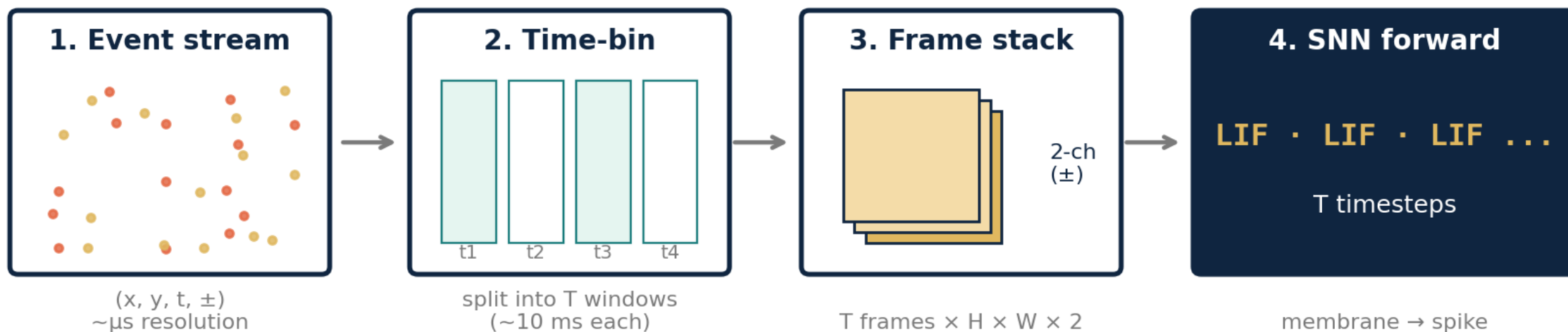
DVS-CIFAR10 (event-based image classification, T = 10).

Dataset	Description	Size
DVS-CIFAR10	CIFAR-10 on monitor + saccading DVS	10 K
N-MNIST	MNIST shown to event camera	70 K
N-Caltech101	Caltech101 in event form	9.1 K
DVS-Gesture (IBM)	11 hand gestures, 29 subjects	1.3 K clips
SHD	Spiking Heidelberg Digits (audio events)	10 K
Gen1 / 1Mpx	Prophesee automotive (object detect.)	39 h / 14 h
ASL-DVS	American Sign Language (event)	100 K

Highlighted: dataset used in this thesis.

From DVS Events to SNN Input

Asynchronous events become T frames that the LIF stack consumes.

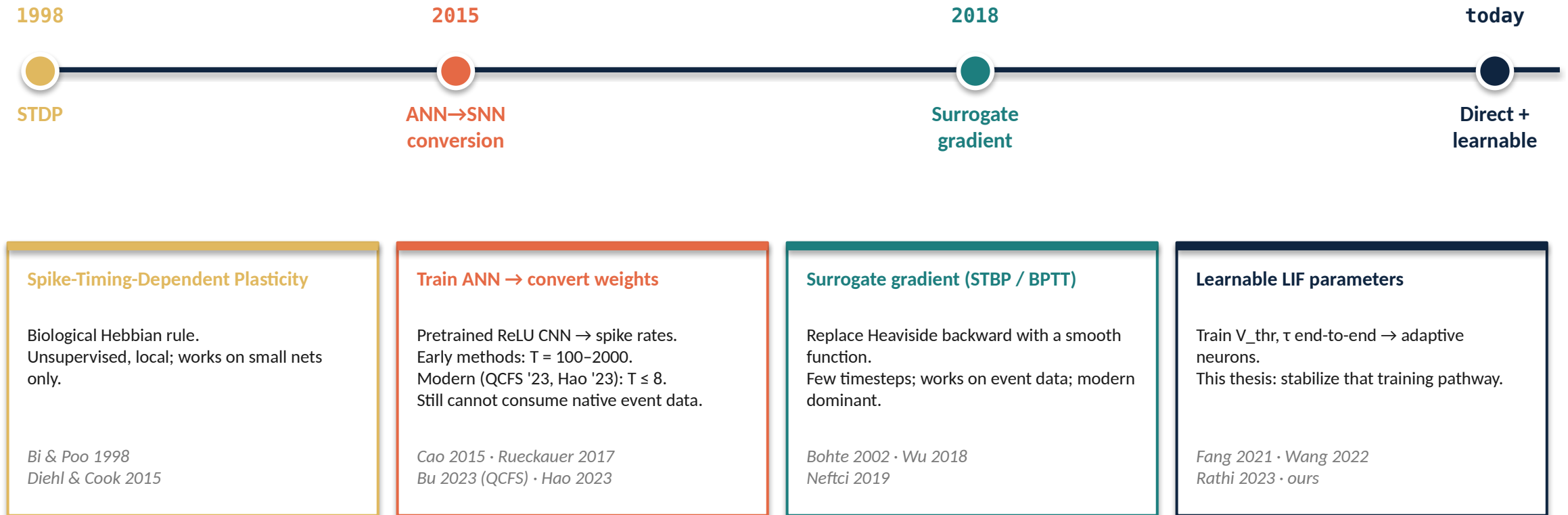


Why this matters for MP-Init

The first frame after binning has no history $\rightarrow$ the membrane starts at zero, far from π . That's exactly the gap MP-Init closes.

How Were SNNs Trained Before?

From biological learning rules to modern surrogate-gradient BPTT.



Where Prior Work Falls Short

TCS methods miss the membrane potential; threshold-learning methods don't diagnose the gradient.

TCS / Normalization in SNNs

Method	What it does	What it misses
BNTT '21	BN on synaptic input current	<i>U[t] drift not corrected</i>
tdBN '20	Threshold-dependent BN on activations	<i>Pre-spike BN; U[t] persists</i>
TEBN '22	Timestep-specific affine on outputs	<i>Membrane drift untouched</i>
TAB '24	Temporal-adaptive output normalization	<i>Membrane TCS persists</i>

Our take (MP-Init)

Fix the cause, not symptom — align $M[t]$ with stationary π directly.

Trainable LIF parameters (V_{thr} , τ)

Method	What's learned	What it misses
PLIF '21	Leakage τ only	<i>V_thr fixed</i>
LTMD '22	$V_{thr} + \tau$ (heavy clipping)	<i>Hides instability</i>
DIET-SNN '23	$V_{thr} + \tau$ (gradient clip)	<i>Doesn't classify failure</i>
Ours (TrSG)	$V_{thr} + \tau$, scale-invariant	—

Our take (TrSG)

Diagnose first (4 failure modes), then cancel the $1/V_{thr}$ factor by construction.